REAL ESTATE PORTAL - DBMS PROJECT

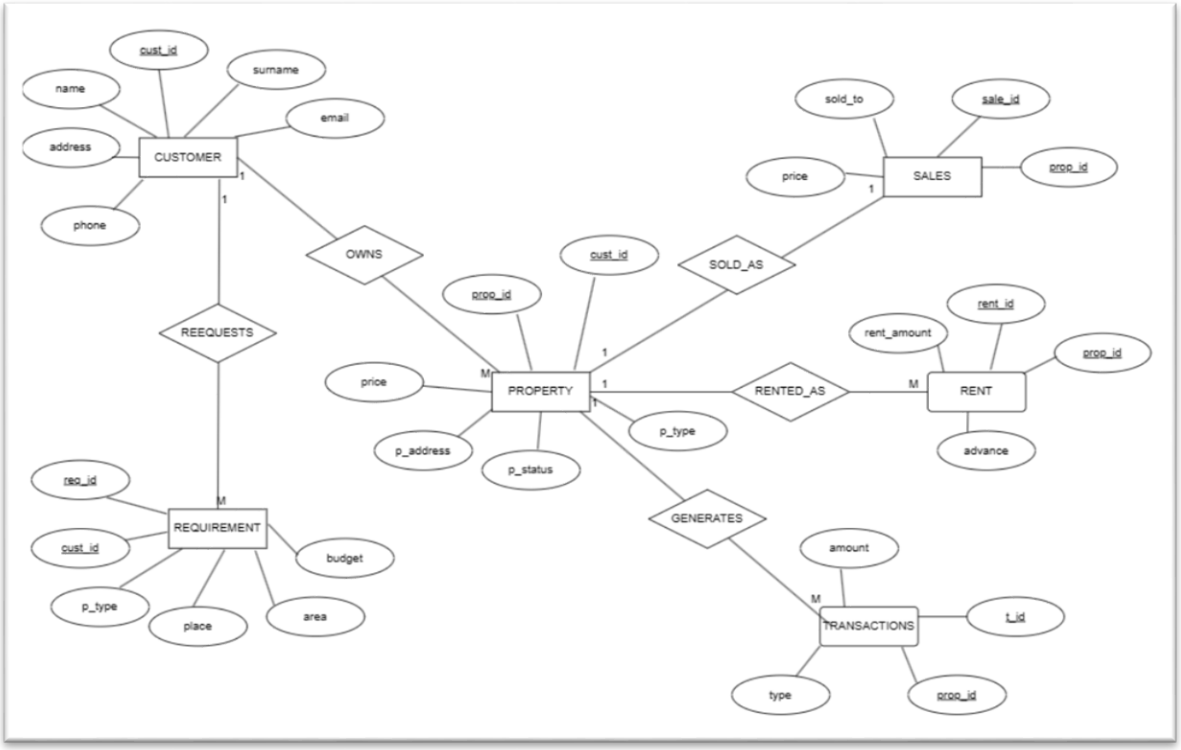
1. Introduction

A database system to manage real estate properties, customers, sales, rentals, and transactions.

2. Objectives

Manage data, maintain integrity, track transactions, support buying/selling.

4. ER Diagram



ER Diagram Description

The ER diagram represents a Real Estate Portal database system consisting of six main entities:

- Customer
- Property
- Sales
- Rent
- Transactions
- Requirements

Each entity contains specific attributes and is connected through well-defined relationships.

Entities and Attributes

1. Customer

Attributes:

1. cust_id (PK)
2. name
3. surname
4. email
5. phone
6. address

❖ The Customer entity represents users of the system, including buyers and sellers.

2. Property

Attributes:

1. prop_id (PK)
2. cust_id (FK)
3. p_type
4. p_status
5. p_address
6. price

❖ The Property entity represents real estate listings associated with customers.

3. Sales

Attributes:

1. sale_id (PK)
2. prop_id (FK)
3. sold_to
4. price

- The Sales entity stores information about properties that have been sold.

4. Rent

Attributes:

1. rent_id (PK)
2. prop_id (FK)
3. rent_amount
4. advance

- The Rent entity contains details of rental agreements for properties.

5. Transactions

Attributes:

1. t_id (PK)
2. prop_id (FK)
3. type
4. amount

- The Transactions entity records all financial transactions related to properties.

6. Requirement

Attributes:

1. req_id (PK)
2. cust_id (FK)
3. p_type
4. place
5. area
6. budget

- The Requirement entity captures customer preferences and property requirements.

Relationships

Customer – Property (Owns / Lists)

A one-to-many (1:M) relationship exists between Customer and Property. A customer can own or list multiple properties, but each property is associated with one customer.

Property – Sales (Sold_As)

A one-to-one (1:1) relationship exists between Property and Sales. Each property can be sold once, resulting in a single sales record.

Property – Rent (Rented_As)

A one-to-many (1:M) relationship exists between Property and Rent. A property can have multiple rental records over time.

Property – Transactions (Generates)

A one-to-many (1:M) relationship exists between Property and Transactions. Each property can generate multiple financial transactions.

Customer – Requirement (Requests)

A one-to-many (1:M) relationship exists between Customer and Requirement. A customer can specify multiple property requirements.

3. Complete SQL (DDL + DML + Queries)

```
CREATE DATABASE real_estate;
USE real_estate;

CREATE TABLE customer (
   cust_id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(50),
   surname VARCHAR(50),
   email VARCHAR(100),
   phone VARCHAR(15),
   address VARCHAR(300)
);

INSERT INTO customer (name, surname, email, phone, address) VALUES
('Ekansh','Verma','ekansh@gmail.com','9876543210','Delhi'),
('Prajjwal','Sharma','prajjwal@gmail.com','9876543211','Mumbai'),
('Tanush','Gupta','tanush@gmail.com','9876543212','Noida'),
('Shivan','Singh','shivan@gmail.com','9876543213','Lucknow'),
('Bora','Kumar','bora@gmail.com','9876543214','Pune'),
('Amit','Yadav','amit@gmail.com','9876543215','Kanpur'),
('Rahul','Mehta','rahul@gmail.com','9876543216','Delhi'),
('Rohit','Verma','rohit@gmail.com','9876543217','Jaipur'),
('Suresh','Patel','suresh@gmail.com','9876543218','Ahmedabad'),
('Vikas','Shah','vikas@gmail.com','9876543219','Surat'),
('Ankit','Jain','ankit@gmail.com','9876543220','Delhi'),
('Deepak','Mishra','deepak@gmail.com','9876543221','Varanasi'),
('Kunal','Agarwal','kunal@gmail.com','9876543222','Agra'),
('Manish','Pandey','manish@gmail.com','9876543223','Lucknow'),
('Nitin','Tyagi','nitin@gmail.com','9876543224','Meerut'),
('Pankaj','Singh','pankaj@gmail.com','9876543225','Patna'),
('Ravi','Kumar','ravi@gmail.com','9876543226','Ranchi'),
('Sanjay','Gupta','sanjay@gmail.com','9876543227','Bhopal'),
('Tarun','Chauhan','tarun@gmail.com','9876543228','Indore'),
('Umesh','Dubey','umesh@gmail.com','9876543229','Gwalior'),
```

```
('Vivek','Saxena','vivek@gmail.com','9876543230','Delhi'),
('Yash','Kapoor','yash@gmail.com','9876543231','Mumbai'),
('Zaid','Ali','zaid@gmail.com','9876543232','Hyderabad'),
('Arjun','Reddy','arjun@gmail.com','9876543233','Chennai'),
('Bhavesh','Joshi','bhavesh@gmail.com','9876543234','Udaipur'),
('Chetan','Rawat','chetan@gmail.com','9876543235','Dehradun'),
('Dev','Bansal','dev@gmail.com','9876543236','Chandigarh'),
('Eshan','Kohli','eshan@gmail.com','9876543237','Delhi'),
('Farhan','Khan','farhan@gmail.com','9876543238','Mumbai');

CREATE TABLE property (
   prop_id INT PRIMARY KEY AUTO_INCREMENT,
   cust_id INT,
   p_type VARCHAR(50),
   p_status VARCHAR(50),
   p_address VARCHAR(200),
   price DECIMAL(10,2),
   FOREIGN KEY (cust_id) REFERENCES customer(cust_id)
);

INSERT INTO property (cust_id, p_type, p_status, p_address, price) VALUES
(1,'Flat','Available','Delhi',5000000),
(2,'House','Sold','Mumbai',12000000),
(3,'Flat','Rent','Noida',3000000),
(4,'Villa','Available','Lucknow',15000000);

CREATE TABLE rent (
   rent_id INT PRIMARY KEY AUTO_INCREMENT,
   prop_id INT,
   rent_amount DECIMAL(10,2),
   advance DECIMAL(10,2),
   FOREIGN KEY (prop_id) REFERENCES property(prop_id)
);

CREATE TABLE requirement (
   req_id INT PRIMARY KEY AUTO_INCREMENT,
   cust_id INT,
   p_type VARCHAR(50),
   place VARCHAR(100),
   area VARCHAR(50),
   budget DECIMAL(10,2),
   FOREIGN KEY (cust_id) REFERENCES customer(cust_id)
);
```

```
CREATE TABLE sales (
   sale_id INT PRIMARY KEY AUTO_INCREMENT,
   prop_id INT UNIQUE,
   sold_to VARCHAR(100),
   price DECIMAL(10,2),
   FOREIGN KEY (prop_id) REFERENCES property(prop_id)
);

CREATE TABLE transactions (
   t_id INT PRIMARY KEY AUTO_INCREMENT,
   prop_id INT,
   type VARCHAR(50),
   amount DECIMAL(10,2),
   FOREIGN KEY (prop_id) REFERENCES property(prop_id)
);

-- Queries

SELECT * FROM customer;
SELECT * FROM property WHERE price > 500000;

SELECT c.name, p.p_type, p.price
FROM customer c
JOIN property p ON c.cust_id = p.cust_id;

SELECT COUNT(*) AS total_properties FROM property;
SELECT AVG(price) FROM property;
SELECT MAX(price) FROM property;
SELECT MIN(price) FROM property;

SELECT p_address, COUNT(*) FROM property GROUP BY p_address;
SELECT * FROM customer WHERE address='Delhi';
SELECT * FROM customer LIMIT 5;
SELECT * FROM customer WHERE name LIKE 'A%';
SELECT * FROM customer WHERE address IN ('Delhi','Mumbai');
SELECT * FROM property WHERE price BETWEEN 3000000 AND 10000000;

SELECT name FROM customer WHERE cust_id IN (
   SELECT cust_id FROM property WHERE price > 5000000
);
```

5. Conclusion

The system successfully models a real estate database with relationships and queries.